Randomized average-case easiness of Sigma2 implies a subexponential algorithm for AM

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Editorial scope. Counted original Theorem36 (Theorem1 Item2) once. Essential hypothesis DistSigma2^P subset of AvgBPP and randomized BPTIME conclusion retained. Full non-routine pKt oracle elimination, witness compression, reconstruction/symmetry and depth arguments included. Original proof accepts yes instances with the stated constant advantage and no instances with at most the stated tiny error; no deterministic or unconditional simulation claimed.

Allowed standard prerequisites (editorial).

- Standard UP/NP and RAM program encodings, direct-product-generator reconstruction [11, Lemma3.14], elementary averaging and Markov bounds. Exact imported DPG Lemma14 retained. All source-local randomized heuristic, amplification, pKt reconstruction, conditional symmetry, unique-witness oracle elimination and depth-telescoping proofs retained. Unused rKt branches are not part of this pKt/UP proof.

2 Preliminaries (necessary author context)

2.1 Notation

We say that a function $p : \mathbb{N} \to \mathbb{N}$ is non-decreasing if $p(\ell) \geq \ell$ for every integer $\ell \geq 0$. Throughout the paper, we will rely on constructions whose associated complexity parameters (e.g., running time) can be bounded by some polynomial p. We will implicitly assume that the polynomials provided in these results are non decreasing in order to avoid including this condition in every statement. This can be done without loss of generality by the monotonicity of our bounds. For a probability distribution D_n, we use $\mathrm{supp}(D_n)$ to denote its support. Given an element x, we let $D_n(x)$ denote the probability of x under D_n. We denote by U_n the uniform distribution over n-bit strings. Let D_1 and D_2 be probability distributions supported over a set X, $A : X \to \{0, 1\}$, and $\varepsilon \geq 0$. We say that A ε-distinguishes D_1 and D_2 if

$$\left| \Pr_{x \sim D_1}[A(x) = 1] - \Pr_{x \sim D_2}[A(x) = 1] \right| \geq \varepsilon.$$

If A is a randomized algorithm, each probability in the expression above also takes into account the internal randomness of A, which will be denoted by r_A.

2.2 Average-case complexity

Recall that a pair $(L, \mathcal{D})$ is a distributional problem if $L \subseteq \{0,1\}^*$ and $\mathcal{D} = \{D_n\}_{n\geq 1}$ is an ensemble of probability distributions, where each D_n is supported over $\{0,1\}^*$. Let $\mathcal{D} = \{D_n\}_{n\geq 1}$ be an ensemble of distributions. We say that $\mathcal{D} \in$ PSamp if there is a randomized polytime algorithm A such that, for every $n \geq 1$, $A(1^n)$ is distributed according to D_n. More generally, we use $D_n \in$ Samp$[T(n)]/a(n)$ to denote that D_n can be sampled by an algorithm that runs in time $T(n)$ and has advice complexity $a(n)$. We let DistNP denote the set of distributional problems $(L, \mathcal{D})$ with $L \in$ NP and $\mathcal{D} \in$ PSamp.

Definition 4 (Avg$_\delta$BPP **[3]).** Let $(L, \mathcal{D})$ be a distributional problem, and $\delta : \mathbb{N} \to [0,1]$. We say that $(L, \mathcal{D}) \in$ Avg$_\delta$BPP if there is randomized polytime algorithm A such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n) = \perp] \geq 1/4\right] \leq \delta(n),$$

where r_A denotes the internal randomness of A. Such an algorithm A is called an randomized errorless heuristic for $(L, \mathcal{D})$ with failure probability at most δ.

Definition 5 (AvgBPP [3]). [1] We say that $(L, \mathcal{D})$ is in AvgBPP if there exist a randomized algorithm A and a polynomial p such that

1. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n,\delta) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n, \delta > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n,\delta) = \perp] \geq 1/4\right] \leq \delta(n),$$
3. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$, $A(x;n,\delta)$ runs in time at most $p(n/\delta)$.

Such an algorithm A is called a randomized errorless heuristic scheme for $(L, \mathcal{D})$.

Lemma 6. Let $(L, \mathcal{D})$ be a distributional problem in Avg$_{n^{-1}}$BPP. There exists a randomized polynomial-time algorithm B such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$, $B(x;n) \in \{0,1\}$,
2. for every $n > 0$, and every $x \in \text{supp}(D_n)$, if $x \in L$, then $\Pr_{r_B}[B(x;n) = 1] \geq 1 - 1/n$, and
3. for every $n > 0$, $\Pr_{x\sim D_n, r_B}[B(x;n) = L(x)] \geq 1 - 3/n$.

[1] Some authors define AvgBPP as $\bigcap_{c>0}$ Avg$_{n^{-c}}$BPP. All our results also hold with respect to that definition.

Proof. Let A be an $\mathrm{Avg}_{n^{-1}}\mathrm{BPP}$ algorithm for $(L, \mathcal{D})$ as in Definition 4. Let A' be the algorithm obtained, using standard amplification techniques,[2] such that for all $x \in \mathrm{supp}(D_n)$,

$$\Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n \tag{1}$$

and

$$\Pr_{x \sim D_n}\left[\Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\right] \leq 1/n. \tag{2}$$

Let B be the randomized algorithm which, on input $x \in \mathrm{supp}(D_n)$, simulates $A'(x;n)$ and outputs 1 if $A'(x;n) \in \{1, \perp\}$ or 0 if $A'(x;n) = 0$. By Equation (1), if $x \in L$, then

$$\Pr_{r_B}[B(x;n) = 0] = \Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n.$$

Let $\mathrm{bad} := \{x \in \mathrm{supp}(D_n) \mid \Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\}$. By Equation (2), at most a $(1/n)$-measure of $x \in \mathrm{supp}(D_n)$ are in bad. Thus,

$$\begin{aligned}
\Pr_{x \sim D_n, r_B}[B(x;n) \neq L(x)] &\leq \Pr_{x, r_{A'}}[A'(x;n) = \perp] + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq \Pr_{x}[x \in \mathrm{bad}] + \Pr_{x, r_{A'}}[A'(x;n) = \perp \mid x \notin \mathrm{bad}] \\
&\quad + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq 1/n + 1/n + 1/n = 3/n.
\end{aligned}$$

This completes the proof. ◁

2.4 Direct product generators (DPGs)

Definition 13 (Direct Product Generator (DPG) [11, Definiton 3.10]). For $k, n \in \mathbb{N}$, we define the k-wise direct product generator to be the function

$$\mathrm{DP}_k : \{0,1\}^n \times \{0,1\}^{nk} \to \{0,1\}^{nk+k}$$

such that

$$\mathrm{DP}_k(x; z^1, \ldots, z^n) := (z^1, \ldots, z^k, x \cdot z^1, \ldots, x \cdot z^k).$$

Lemma 14 (DPG Reconstruction [11, Lemma 3.14]). For any $n, k \in \mathbb{N}$ with $k \leq 2n$, and $\varepsilon > 0$, there exists a pair of algorithms A and $\mathrm{Recon}^{(-)}$ such that

- Recon^{D} takes oracle access to an oracle $D : \{0,1\}^{nk+k} \to \{0,1\}$.
- $A : \{0,1\}^n \times \{0,1\}^r \to \{0,1\}^k$ is called the advice function and is computable in time $\mathrm{poly}(n/\varepsilon)$.
- $\mathrm{Recon}^{(-)} : \{0,1\}^k \times \{0,1\}^r \to \{0,1\}^n$ is called a reconstruction procedure and is computable in time $\mathrm{poly}(n/\varepsilon)$.[3]

[2] In more detail, A' can be obtained from A by running it $O(n)$ times and selecting the most common output.

[3] We note that if the oracle is a uniform algorithm that runs in time $t(m)$ on inputs of length m, the reconstruction procedure can be computed in time $t(nk+k) \cdot \mathrm{poly}(n/\varepsilon)$.

- The randomness complexity r is at most $\text{poly}(n/\varepsilon)$.
- For any string $x \in \{0,1\}^n$ and any function D that ε-distinguishes $\text{DP}_k(x; U_{nk})$ from U_{nk+k}, it holds that

$$\Pr_{w \sim \{0,1\}^r}[\text{Recon}^D(A(x,w),w) = x] \geq 1/\text{poly}(n/\varepsilon).$$

2.7 Encoding tuples (author context)

We will also need to consider distributions that generate input instances that are tuples, while the only known information to the distributions is the length of the instances. To do this, we will encode the format as the length of the instances. More specifically, we use the following way to encode tuples [3].

Proposition 16 ([3]; see also [6, Proposition 3.1]). There is a pair of polytime encoding-decoding algorithms $\text{Enc} : \mathbb{N}^* \to \mathbb{N}$ and $\text{Dec} : \mathbb{N} \to \mathbb{N}^* \cup \{\perp\}$ such that

1. For $k \in \mathbb{N}$ and k integers $n_1, n_2, \ldots, n_k$, we write $\langle n_1, n_2, \ldots, n_k \rangle := \text{Enc}(n_1, n_2, \ldots, n_k)$, and we have
   (a) $\sum_{i=1}^{k}(2 \cdot (\lfloor \log n_i \rfloor + 2) + n_i) \leq \langle n_1, \ldots, n_k \rangle \leq O_k(\prod_{i=1}^{k}(n_i \cdot \log^2 n_i))$.
   (b) $\text{Dec}(\langle n_1, n_2, \ldots, n_k \rangle) = n_1, \ldots, n_k$. Here, Dec takes the binary representation of an integer and outputs k integers which are also represented in binary.
2. $\text{Dec}(u) = \perp$ if $u \neq \langle \vec{n} \rangle$ for any $\vec{n} \in \mathbb{N}^*$.

Proof. Given k integers $n_1, n_2, \ldots, n_k$, the encoding algorithm Enc first obtains the binary string

$$z := \alpha_1 \circ 01 \circ \alpha_2 \circ 01 \circ \alpha_k \circ 01 \circ \text{bin}(n_1) \circ \text{bin}(n_2) \circ \cdots \circ \text{bin}(n_k),$$

where for each $i \in [k]$, α_i is obtained from $\text{bin}(|\,\text{bin}(n_i)|)$ with every bit doubled. Then Enc outputs the integer n whose binary representation is z. It is easy to see that given the binary representation of n, which is simply the string z, one can recover the $n_1, n_2, \ldots, n_k$ in time polynomial in $|z|$. Also, item (a) can be verified by a simple calculation. ◁

3 Probabilistic Time-Bounded Kolmogorov Complexity (author excerpts)

In this section, we formally define probabilistic Kolmogorov complexity and prove some useful properties of this notion, which will be used later in proving our main results.

Definition 17 (pK^t). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N}$, an oracle A, and $\delta \in [0,1]$, the δ-probabilistic A-oracle t-bounded Kolmogorov complexity of x given y is defined as

$$\mathrm{pK}^{t,A}_\delta(x \mid y) := \min\left\{k \;\middle|\; \Pr_{w\sim\{0,1\}^t}[\exists M \in \{0,1\}^k, M^A(w,y) \text{ outputs } x \text{ within } t \text{ steps}] \geq \delta\right\},$$

where M is a RAM-machine.[4] We omit the subscript δ if $\delta = 2/3$, omit the superscript A if $A = \emptyset$, omit "$\mid y$" if y is the empty string, and omit the superscript t if $t = \infty$. For simplicity, in the rest of this section we only consider pK^t without oracles. It is easy to see that all the results also hold with any oracle.

Lemma 21 (Success Amplification). For any string $x \in \{0,1\}^n$, time bound $t \in \mathbb{N}$, and $0 \leq \alpha < \beta \leq 1$, we have

$$\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq \mathrm{pK}^t_\alpha + O(\log(q/\alpha)),$$

where $q = \ln(1/(1-\beta))$.

Proof. Suppose $\mathrm{pK}^t_\alpha(x) = k$. Then by definition, we have

$$\Pr_{w\sim\{0,1\}^t}[\exists M_w \in \{0,1\}^k \text{ such that } M_w(w) \text{ outputs } x \text{ within } t \text{ steps}] \geq \alpha.$$

Call such a w good. After sampling ℓ independent $w^{(1)}, w^{(2)}, \ldots, w^{(\ell)} \in \{0,1\}^t$, the probability that no $w^{(i)}$ is good is at most $(1-\alpha)^\ell \leq e^{-\alpha\ell}$, which can be made at most $1-\beta$ by choosing $\ell = q/\alpha$. It follows that, with probability at least β over a random $w = (w^{(1)}, w^{(2)}, \ldots, w^{(q/\alpha)}) \in \{0,1\}^{qt/\alpha}$, there exists an index $1 \leq i \leq q/\alpha$ (described with at most $\lceil \log(q/\alpha) \rceil$ bits) and a program $M_{w^{(i)}} \in \{0,1\}^k$ such that $M_{w^{(i)}}(w^{(i)})$ outputs x within $O(qt/\alpha)$ steps. Hence, $\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq k + O(\log(q/\alpha))$. ◁ Note that the parameter α affects the time complexity in the resulting $\mathrm{pK}^{O(qt/\alpha)}_\beta$ bound. Lemma 21 is sufficient in applications where $\alpha \geq 1/\operatorname{poly}(n)$.

3.2 Bounding pK^t via DPG reconstruction (selected original case)

The following is a key lemma for pK^t that we will use in proving our main results related to randomized average-case complexity.

Lemma 22 (pK^t Reconstruction Lemma). For $\varepsilon > 0$, $x \in \{0,1\}^n$, $s \in \mathbb{N}$, and $k \in \mathbb{N}$ satisfying $k \leq 2n$, let D be a randomized algorithm that takes an advice string β and runs in time t_D such that D ε-distinguishes $\mathrm{DP}_k(x; U_{nk})$ from U_{nk+k}. Then there is a polynomial p_{DP} such that

$$\mathrm{pK}^{\widetilde{O}(t_D)\cdot p_{\mathrm{DP}}(n/\varepsilon)}(x \mid \beta) \leq k + \log p_{\mathrm{DP}}(nt_D/\varepsilon).$$

[4] We use the RAM model in this definition for convenience: it simplifies the time bound estimates for a machine M that takes as input both randomness w and an "auxiliary" string y, and allows both w and y to be as long as the running time of the RAM-machine M; cf. [29] for a similar definition in the case of deterministic time-bounded conditional Kolmogorov complexity. Alternatively, we could use a TM model, with inputs w and y presented on two different tapes. However, we prefer the RAM model as it's also useful in the context of fine-grained worst-case to average-case reductions [6].

Proof. We will view the distinguisher D as a deterministic algorithm that takes a string of t_D bits, denoted by r_D, as its internal randomness. By our assumption (and dropping the absolute value sign without loss of generality), we have

$$\mathbb{E}_{y\sim\{0,1\}^{nk},r_D\sim\{0,1\}^{t_D}}[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_{z\sim\{0,1\}^{nk+k},r_D\sim\{0,1\}^{t_D}}[D(z;r_D)]\geq\varepsilon.$$

Then by an averaging argument, we have

$$\Pr_{r_D}[\mathbb{E}_y[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_z[D(z;r_D)]\geq\varepsilon/2]\geq\varepsilon/2. \tag{4}$$

In other words, with probability at least $\varepsilon/2$ over the internal randomness r_D, $D(-,r_D)$ is an $(\varepsilon/2)$-distinguisher for $\mathrm{DP}_k(x;U_{nk})$ and U_{nk+k}. Let us say that r_D is good if this is true. Let $\mathrm{Recon}^{(-)}:\{0,1\}^k\times\{0,1\}^r\to\{0,1\}^n$ be the reconstruction procedure in Lemma 14 that works for distinguishing parameter $\varepsilon/2$, where the randomness complexity r is $\mathrm{poly}(n/\varepsilon)$. We have

$$\begin{aligned}
&\Pr_{w\sim\{0,1\}^r,r_D\sim\{0,1\}^{t_D}}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x]\\
&\geq \Pr_{w,r_D}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x \mid r_D \text{ is good}]\cdot\Pr_{r_D}[r_D \text{ is good}]\\
&\geq \frac{1}{\mathrm{poly}(n/\varepsilon)}\cdot(\varepsilon/2). \qquad \text{(Lemma 14 and Equation (4))}
\end{aligned}$$

The above implies that for at least $1/\mathrm{poly}(n/\varepsilon)$ fraction of the randomness (w,r_D), there exists a program M, which takes k bits for some α (that can depend on the randomness) and the advice of β used by D, such that M simulates $\mathrm{Recon}^{D(-,r_D)}$ on (α,w) and outputs x. Note that since Recon^D runs in time $t_D\cdot\mathrm{poly}(n/\varepsilon)$, our program M can be made to run in time $T:=\widetilde{O}(t_D)\cdot\mathrm{poly}(n/\varepsilon)$. Therefore, we have

$$\mathrm{pK}^T_{1/\mathrm{poly}(n/\varepsilon)}(x\mid\beta)\leq k+O(\log(nt_D/\varepsilon)).$$

Then the lemma follows easily from Lemma 21. ◁

3.4 Symmetry of information (original Item1 and its full proof)

Lemma 26 (Symmetry of information for pK^t and rK^t under the average-case easiness of NP), Item 1. If DistNP $\subseteq$ AvgBPP, then there exist polynomials p and p_0 such that for all sufficiently large $n,m\in\mathbb{N}$, all $t\geq p_0(n,m)$, and all $0\leq\tau\leq t$ the following holds: for every $x\in\{0,1\}^n$ and every family $\{y_r\in\{0,1\}^m\mid r\in\{0,1\}^\tau\}$, with probability at least $9/10$ over $r\in\{0,1\}^\tau$,

$$\mathrm{pK}^t(x,y_r\mid r)>\mathrm{pK}^{p(t)}(x\mid r)+\mathrm{pK}^{p(t)}(y_r\mid x,r)-\log p(t).$$

Moreover, the special case without conditioning on r also holds for both items above, i.e., when r is the empty string (for $\tau=0$) and y is an arbitrary single string.

Proof of Item 1. Define a language

$$L := \{(u, v, w, w', 1^s) \mid \exists M \in \{0,1\}^s, M(w, w') \text{ prints } uv \text{ in } |w| \text{ steps, and } s = |u|+|v|-10\}.$$

Note that $L \in$ NP. Define a distribution family $\mathcal{D} = \{D_{\langle nk+k, mk'+k', 2t, \tau, s\rangle}\}$ as follows: sample $u \sim U_{nk+k}$, $v \sim U_{mk'+k'}$, $w \sim U_{2t}$, and $w' \sim U_\tau$, and then output $(u, v, w, w', 1^s)$. Note that the ensemble $\mathcal{D}$ is in PSamp under an appropriate encoding of its defining tuple (for simplicity, we omit the padding in the output of $\mathcal{D}$; see Section 2.1). Using the assumption that DistNP $\subseteq$ AvgBPP, it follows that $(L, \mathcal{D}) \in$ AvgBPP. Let B be a randomized algorithm for $(L, \mathcal{D})$ as in Lemma 6. Let $x \in \{0,1\}^n$ and $\{y_r\}_r \subseteq \{0,1\}^m$ be given (for sufficiently large n and m). Let $k, k' \in \mathbb{N}$ be arbitrary parameters such that $k \leq n$ and $k' \leq m$. Observe that there exists a polynomial p_0 such that for any $t \geq p_0(n, m)$, some constant d, and all $z \in \{0,1\}^{nk}$, $z' \in \{0,1\}^{mk'}$, and $r \in \{0,1\}^\tau$,

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) \leq \mathrm{pK}^t(x, y_r \mid r) + |z| + |z'| + d \log t. \tag{6}$$

Here $p_0(n, m)$ reflects the time required to deterministically compute $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$ given x, y_r, z, z', and $d \log t$ bits of information to delineate x from y_r. Let $t \geq p_0(n, m)$. By a counting argument, for independently random u, v, w, and w',

$$\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \leq \frac{2^s \cdot 2^{|w|+|w'|}}{2^{|u|+|v|+|w|+|w'|}} = 2^{s-|u|-|v|} = 2^{-10},$$

where the last equality is by the definition of L requiring that $s = |u| + |v| - 10$. Then

$$\begin{aligned}\Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) = 1] \leq &\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \\ &+ \Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) \neq L(u, v, w, w', 1^s)] \\ &\leq 2^{-10} + (3/n). \end{aligned} \tag{7}$$

By Markov's inequality, for at least 9/10 fraction of random strings $r \in \{0,1\}^\tau$, we get

$$\Pr_{u,v,w,r_B}[B(u, v, w, r, 1^s) = 1] \leq 10 \cdot (2^{-10} + (3/n)) \leq 1/10. \tag{8}$$

Fix any such r so that Equation (8) holds. This also fixes the string y_r. Next we show, using a hybrid argument, that $B(-, U_{2t}, r, 1^s)$ cannot distinguish between the uniform distribution and the distribution $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$, for random independent z, z', where $k \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(x \mid r)$ and $k' \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(y_r \mid x, r)$. This will imply that

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] < 2/3$$

for some z, z', yielding the desired lower bound on $\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r)$ via our choice of s and the definition of L. We give the details of the hybrid argument next. First, toward a contradiction, suppose

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] > 1/2.$$

In this case, comparing with Equation (8), we get a randomized distinguisher (with advice) for $\mathrm{DP}_k(x; U_{nk})$, defined by sampling $v \sim U_{mk'+k'}$ and $w \sim U_{2t}$, and outputting $B(-, v, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q(t)}(x \mid r) \leq k + \log q(t) \tag{9}$$

for some polynomial q such that $q(t) \geq p_{\mathrm{DP}}(t_B \cdot 3 \cdot n)$ whenever $t \geq n + m$ and $s \leq n^3$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, v, w, r, 1^s)$. Define $k := \mathrm{pK}^{q(t)}(x \mid r) - \log q(t) - 1$ so that Equation (9) does not hold. Assume for now that $k > 0$. Hence,

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] \leq 1/2. \tag{10}$$

Again, toward a contradiction, suppose that for all z, z',

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \geq 2/3.$$

By definition of B, this implies that

$$\Pr_{z,z',w,r_B}[B(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) = 1] \geq (2/3)(1 - 1/n) > 5/8.$$

In this case, comparing with Equation (10), we get a randomized distinguisher (with advice) B' for $\mathrm{DP}_{k'}(y_r; U_{mk'})$, defined by sampling $z \sim U_{nk}$, $w \sim U_{2t}$, and outputting $B(\mathrm{DP}_k(x; z), -, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q'(t)}(y_r \mid x, r) \leq k' + \log q'(t) \tag{11}$$

for some polynomial q' with $q'(t) \geq p_{\mathrm{DP}}(t_{B'} \cdot 8 \cdot m)$ whenever $t \geq n + m$ and $s \leq n^3$, where $t_{B'}$ denotes the time required to compute B'. We now choose $k' := \mathrm{pK}^{q'(t)}(y_r \mid x, r) - \log q'(t) - 1$ so that Equation (11) does not hold. Assume for now that $k' > 0$. Hence, there exist z and z' such that

$$\begin{aligned}&\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \\ &= \Pr_w[\exists M \in \{0,1\}^s, M(w, r) \text{ outputs } (\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z')) \text{ within } |w| \text{ steps}] < 2/3,\end{aligned}$$

which implies that

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > s.$$

For this choice of z, z', by definition of s we get that $s = |u| + |v| - 10 = |z| + k + |z'| + k' - 10$, and so

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > |z| + k + |z'| + k' - 10.$$

Combining this inequality with Equation (6), we get

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) &\geq \mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) - |z| - |z'| - d \log t \\ &> k + k' - d \log t - 10.\end{aligned} \tag{12}$$

The definitions of k and k' then imply that

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{q(t)}(x \mid r) + \mathrm{pK}^{q'(t)}(y_r \mid x, r) \\ - \log q(t) - \log q'(t) - d \log t - 12.\end{aligned}$$

For the polynomial $p(t) := (t^{d+1}) \cdot q(t) \cdot q'(t)$ and for every $t \geq p_0(|x|, |y|)$, we get

$$\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{p(t)}(x \mid r) + \mathrm{pK}^{p(t)}(y_r \mid x, r) - \log p(t),$$

as desired. Finally, consider the case that $k \leq 0$ or $k' \leq 0$. If $k \leq 0$, then $\mathrm{pK}^{q(t)}(x \mid r) \leq \log q(t) + 1$, implying that $\mathrm{pK}^{p(t)}(x \mid r) < \log p(t)$. But then the lemma simply follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(y_r \mid x, r)$. Similarly, if $k' \leq 0$, then $\mathrm{pK}^{p(t)}(y_r \mid x, r) < \log p(t)$, and the lemma follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(x \mid r)$. ◁

3.7 Probabilistic computational depth (original Item1 and its full proof)

The next lemma is a straightforward adaptation of an argument from [11]. We provide a proof for completeness.

Lemma 31 (Computational Depth Upper Bound), Item 1. For all $x \in \{0,1\}^n$ and $r \in \{0,1\}^*$, and for all non-decreasing polynomials p_0 and p, if n is large enough then there exists a time bound t_1 such that $p_0(n) \leq t_1 \leq 2^{n/\log n}$ and

$$\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) \leq O(n/\log n).$$

Proof. We will start by proving Item 1. Given $x \in \{0,1\}^n$ and polynomials p_0 and p, define the polynomial $\tau := p \circ p_0$. For an integer $I \geq 1$, consider the following telescoping sum:

$$\begin{aligned}\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r) = {} & (\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^2(n)}(x \mid r)) \\ & + (\mathrm{pK}^{\tau^2(n)}(x \mid r) - \mathrm{pK}^{\tau^3(n)}(x \mid r)) \\ & + \cdots + (\mathrm{pK}^{\tau^I(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r)),\end{aligned}$$

where $\tau^i(-)$ denotes the composition of τ with itself i times. For any choice of x, p_0, and p as in the statement of the lemma, $\mathrm{pK}^{\tau(n)}(x \mid r) \leq n + d$, for some universal constant $d \geq 0$; hence, the above sum is at most $n + d$. By averaging, there is some index $i_0 \in [I]$ such that

$$\mathrm{pK}^{\tau^{i_0}(n)}(x \mid r) - \mathrm{pK}^{\tau^{i_0+1}(n)}(x \mid r) \leq (n+d)/I.$$

For this i_0, define $t_1 := \tau^{i_0}(n)$. Note that $t_1 \geq \tau(n) \geq p_0(n)$, since $i_0 \geq 1$ and $p(\ell) \geq \ell$ for every input ℓ. Letting $c \in \mathbb{N}$ be such that $\tau(n) \leq n^c$ for sufficiently large n, define $I := \log_c(n/(\log_2 n)^2)$. Then $t_1 \leq n^{c^I} = 2^{n/\log n}$. Moreover,

$$\begin{aligned}\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) &\leq \mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{\tau(t_1)}(x \mid r) \\ &\leq O(n/\log n),\end{aligned}$$

as desired.

4 Probabilistic Worst-Case to Average-Case Reductions (AM case)

4.1 Auxiliary lemmas

Given a language $L \in \mathrm{NP}$ with a corresponding verifier V, and $x \in L$, let y_x be the lexicographically first witness that $x \in L$. It is not hard to compute y_x from x if an NP oracle is available, by performing a standard search-to-decision reduction. For this reason, the time-bounded Kolmogorov complexity of the pair (x, y_x) is essentially that of x, in the presence of an NP oracle. We will show, more generally, that the oracle can be eliminated under an appropriate average-case easiness assumption, even for the following randomized versions of Σ_ℓ^{P}, denoted $\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$.

Definition 32 ($\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$). A language L is in the class $\mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$ if there is a $\mathrm{BP} \circ \Sigma_\ell \circ \mathrm{P}$ formula

$$\varphi(x) = \mathrm{BP}_r \exists y_1 \forall y_2 \ldots Q y_\ell \; R(x, y_1, y_2, \ldots, y_\ell, r)$$

for a polytime predicate R and all string variables $y_1, \ldots, y_\ell, r$ of length $\mathrm{poly}(|x|)$, such that, for all $x \in \{0,1\}^n$,

$$x \in L \implies \Pr_r[\exists y_1 \forall y_2 \ldots Q y_\ell \; R(x, y_1, y_2, \ldots, y_\ell, r)] \geq \delta, \text{ and}$$
$$x \notin L \implies \Pr_r[\forall y_1 \exists y_2 \ldots \overline{Q} y_\ell \; \neg R(x, y_1, y_2, \ldots, y_\ell, r)] \geq \delta.$$

Lemma 33 (Oracle Elimination), original Item 1. For any $\ell \geq 1$, suppose $\mathrm{Dist}\Sigma_{\ell+1}^{\mathrm{P}} \subseteq \mathrm{AvgBPP}$. For an arbitrary $L \in \mathrm{BP}_\delta \circ \Sigma_\ell^{\mathrm{P}}$, let $x \in L_n$ be sufficiently large, let $r \in \{0,1\}^{n^c}$ be any random string under the BP quantifier such that there exists a witness for the left-most $\exists$ quantifier in the definition of L, and let $y_{x,r} \in \{0,1\}^{n^c}$ be the lexicographically first such L-witness for x and randomness r, for some constant $c > 0$. There exist polynomials q and q_0 such that, for all sufficiently large $n \in \mathbb{N}$, all $x \in L_n$, and all $t \geq q_0(n)$, we have with probability at least $\delta - 1/10$ over $r \in \{0,1\}^{n^c}$ that an L-witness $y_{x,r}$ exists, and

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t).$$

Moreover, for the case of no BP quantifier, i.e., for $L \in \Sigma_\ell^{\mathrm{P}}$, we get the same conclusions (with probability 1) without conditioning on r.

Proof of Item 1. Define

$$L' := \{(\mathrm{DP}_k(x, y; z), w, w', 1^k, 1^s, 1^n) \mid \exists M \in \{0,1\}^s,$$
$$M^{\Sigma_\ell^{\mathrm{P}}}(w, w') \text{ prints } (x, y) \in \{0,1\}^{n+n^c} \text{ within } |w| \text{ steps, where } |w'| = n^c, \text{ and } s = k - 10\}.$$

More formally, in this and subsequent proofs, the above notation should be interpreted as follows: a string $(u, w, w', 1^k, 1^s, 1^n)$ belongs to L' iff there exist some $(x, y) \in \{0,1\}^{n+n^c}$ and $M \in \{0,1\}^s$ such that $\mathrm{DP}_k(x, y; z) = u$, where $u = (z, \alpha)$ for some $z \in \{0,1\}^{(n+n^c)k}$ and $\alpha \in \{0,1\}^k$, and $M^{\Sigma_\ell^{\mathrm{P}}}(w, w')$ outputs (x, y) within $|w|$ steps, for $|w'| = n^c$. Note that $L' \in \mathrm{NP}^{\Sigma_\ell^{\mathrm{P}}}$. Define a distribution

family $\mathcal{D} = \{D_{\langle (n+n^c)k+k, 2t, n^c, k, s, n\rangle}\}$ as follows: sample $u \sim U_{(n+n^c)k+k}$, $w \sim U_{2t}$, $w' \sim U_{n^c}$, and output $(u, w, w', 1^k, 1^s, 1^n)$, for $s = k - 10$. By assumption, $(L', \mathcal{D}) \in \mathrm{AvgBPP}$. Let B be a randomized algorithm for $(L', \mathcal{D})$ as in Lemma 6. For all (sufficiently large) n, k, and s, and for independent uniformly random u, w, and w',

$$\begin{aligned}
&\Pr_{u,w,w'}[(u, w, w', 1^k, 1^s, 1^n) \in L'] \\
&= \Pr_{u,w,w'}[\exists (x, y) \in \{0,1\}^{n+n^c}, \exists z \in \{0,1\}^{(n+n^c)k}, \exists M \in \{0,1\}^s, \\
&\qquad M^{\Sigma_\ell^{\mathrm{P}}}(w, w') = (x, y) \wedge u = \mathrm{DP}_k(x, y; z)] \\
&\leq \frac{2^{s+|w|+|w'|+|z|}}{2^{k+|w|+|w'|+|z|}} = 2^{s-k} = 2^{-10},
\end{aligned}$$

where the inequality is by a union bound and a counting argument, and the last equality by the condition that $s = k - 10$ in the definition of L'. Hence,

$$\begin{aligned}
\Pr_{u,w,w',r_B}[B(u, w, w', 1^k, 1^s, 1^n) = 1] &\leq \Pr_{u,w,w'}[(u, w, w', 1^k, 1^s, 1^n) \in L'] \\
&\quad + \Pr_{u,w,w',r_B}[B(u, w, w', 1^k, 1^s, 1^n) \neq L'(u, w, w', 1^k, 1^s, 1^n)] \\
&\leq 2^{-10} + (3/n).
\end{aligned}$$

By Markov's inequality, for at least $9/10$ fraction of strings r,

$$\Pr_{u,w,r_B}[B(u, w, r, 1^k, 1^s, 1^n) = 1] \leq 10 \cdot (2^{-10} + (3/n)) \leq 1/10. \tag{18}$$

Let $x \in L_n$ be arbitrary. By definition of the BP_δ quantifier, for at least δ fraction of random strings $r \in \{0,1\}^{n^c}$, there exists an L-witness, and hence, the lexicographically first L-witness $y_{x,r}$. By the above, for at least $\delta - 1/10$ of random r, we have that both a witness $y_{x,r}$ exists and Equation (18) holds. Fix any such r. Observe that for some polynomial q_0 (dependent on L) and some constant d, for any $t \geq q_0(n)$,

$$\mathrm{pK}^{2t, \Sigma_\ell^{\mathrm{P}}}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + d \log n =: s(x, r). \tag{19}$$

In particular, $q_0(n)$ reflects the time required to deterministically compute $y_{x,r}$, given x and r, by a search-to-decision procedure for L using a Σ_ℓ^{P}-oracle. Define $s := s(x, r)$ (which determines $k = s + 10$), and let $t \geq q_0(n)$. By the definitions of $\mathrm{pK}^{2t, \Sigma_\ell^{\mathrm{P}}}$ and L', for every $z \in \{0,1\}^{(n+n^c)k}$,

$$\Pr_w[(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) \in L'] \geq 2/3.$$

Then by the definition of B,

$$\begin{aligned}
&\Pr_{z,w,r_B}[B(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) = 1] \\
&\geq \Pr_{z,w}[(\mathrm{DP}_k(x, y_{x,r}; z), w, r, 1^k, 1^s, 1^n) \in L'] \cdot \Pr_{r_B}[B(\omega) = L'(\omega) \mid \omega \in L'] \\
&\geq \frac{2}{3}(1 - (1/n)) \geq 2/3 - o(1).
\end{aligned} \tag{20}$$

Comparing Equation (18) with Equation (20), it is clear that $B(-, U_{2t}, r, 1^k, 1^s, 1^n)$ is a randomized algorithm (with advice) that (1/2)-distinguishes $\mathrm{DP}_k(x, y_{x,r}; U_{(n+n^c)k})$ from uniform. Lemma 22 implies that

$$\begin{aligned} \mathrm{pK}^{p'(t)}(x, y_{x,r} \mid r, s, n, t) &\leq k + \log p'(t) \\ &= s + 10 + \log p'(t) \\ &= \mathrm{pK}^t(x \mid r) + d \log n + \log p'(t) + 10, \end{aligned} \tag{21}$$

for some polynomial p' with $p'(t) \geq p_{\mathrm{DP}}(t_B \cdot 3 \cdot (n + n^c))$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, U_{2t}, r, 1^k, 1^s, 1^n)$. As the advice (s, n, t) can be encoded with $\log s + \log n + \log t + O(\log \log n) \leq (3.1) \cdot \log t$ bits, it follows that $\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t)$ for the polynomial $q(t) := t^{d+4} \cdot p'(t)$. The "Moreover" statement follows the same argument, dropping any mention of w' (and r). ◁

Lemma 34 (Witness Compression), original Item 1. For any $\ell \geq 1$, suppose $\mathrm{Dist}\Sigma^{\mathrm{P}}_{\ell+1} \subseteq \mathrm{AvgBPP}$. For an arbitrary $L \in \mathrm{BP}_\delta \circ \Sigma^{\mathrm{P}}_\ell$, let $x \in L_n$ be sufficiently large, $r \in \{0,1\}^{n^c}$ be any good random string, and let $y_{x,r} \in \{0,1\}^{n^c}$ be the lexicographically first L-witness for x with respect to randomness r, for some constant $c > 0$. Let p_0, p and q_0, q be the polynomials from Items 1 of Lemmas 26 and 33 respectively. Then for every $t \geq \max\{p_0(n), q_0(n + n^c)\}$, with probability at least $\delta - 1/5$ over r, an L-witness $y_{x,r}$ exists, and

$$\mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) \leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r)) + 2 \log p(q(t)).$$

Moreover, for $L \in \Sigma^{\mathrm{P}}_\ell$, we get the same conclusions without conditioning on r.

Proof. We start by proving Item 1. By Item 1 of Lemma 33 and Lemma 26, we get by a union bound that, with probability at least $\delta - 2/10$ over r,

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) \leq \mathrm{pK}^t(x \mid r) + \log q(t),$$

and

$$\mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) > \mathrm{pK}^{p(q(t))}(x \mid r) + \mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) - \log p(q(t)).$$

Combining the previous two inequalities,

$$\begin{aligned} \mathrm{pK}^{p(q(t))}(y_{x,r} \mid x, r) &< \mathrm{pK}^{q(t)}(x, y_{x,r} \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r) + \log p(q(t)) \\ &\leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{p(q(t))}(x \mid r)) + 2 \log p(q(t)). \end{aligned}$$

The "Moreover" part follows by applying the "no random r" versions of Lemma 33 and Lemma 26.

4.2 Case of AM

Theorem 36. If $\mathrm{Dist}\Sigma^{\mathrm{P}}_2 \subseteq \mathrm{AvgBPP}$, then $\mathrm{AM} \subseteq \mathrm{BPTIME}[2^{O(n/\log n)}]$.

Proof. Consider an arbitrary $L \in \mathrm{AM}$ given by a $\mathrm{BP} \circ \Sigma_1^{\mathrm{P}}$ formula such that, for every $x \in \{0,1\}^n$,

$$x \in L \implies \mathrm{BP}_{(1-2^{-n})} r \; \exists y \; V(x,y,r),$$
$$x \notin L \implies \mathrm{BP}_{(1-2^{-n})} r \; \forall y \; \neg V(x,y,r),$$

for a polytime verifier $V(x,y,r)$, where $|r|, |y| \leq n^c$ for some constant $c > 0$, and the notation $\mathrm{BP}_{(\delta)} r \; \varphi(-,r)$, for a formula φ, means $\Pr_r[\varphi(-,r)] \geq \delta$. For a given (sufficiently large) $x \in L_n$, and a good random string r, let $y_{x,r}$ be the lexicographically first L-witness for x, r. By Lemma 34, for some constant $a > 0$, and for all large enough time bounds t, we have, with probability at least $1 - 2^{-n} - 1/5$ over random strings r, that

$$\mathrm{pK}^{t^a}(y_{x,r} \mid x, r) \leq (\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{t^a}(x \mid r)) + (2a) \log t.$$

By Lemma 31, there exists a time bound $t \leq 2^{O(n/\log n)}$ such that $\mathrm{pK}^t(x \mid r) - \mathrm{pK}^{t^a}(x \mid r) \leq bn/\log n$, for some universal constant $b > 0$. Hence, for some universal constant $d > 0$ (independent of x, r),

$$\mathrm{pK}^{2^{dn/\log n}}(y_{x,r} \mid x, r) \leq dn/\log n.$$

A randomized algorithm for L is now obvious:

- Given $x \in \{0,1\}^n$, sample uniformly random strings $r \in \{0,1\}^{|x|^c}$ and $w \in \{0,1\}^{2^{dn/\log n}}$, and then exhaustively search over all $dn/\log n$-bit machines M, running each $M(w,x,r)$ for $2^{dn/\log n}$ steps to produce a candidate witness y, accepting if $V(x,y,r)$ accepts.

The runtime $2^{O(n/\log n)}$ is clear. For correctness, for every $x \in L_n$, we know there is at least a $(4/5 - 2^{-n})$ fraction of good random strings r for which a witness $y_{x,r}$ will be found by our algorithm, with probability at least $2/3$ over random w's (by the definition of pK). So our algorithm accepts $x \in L$ with probability at least $(2/3)(4/5) - o(1) > 1/2$. On the other hand, for any $x \notin L$, our algorithm may find a witness y and accept x for at most 2^{-n} fraction of r's. ◁

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